

Solutions for Graduate Combinatorics HW

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Due to limited time the following are only **sketches** of full solutions, and in particular these solutions alone wouldn't necessarily constitute a solution worth full marks. I also emphasize that there may exist other (possibly simpler) solutions these problems.

1 HW1

- 1.1 Let $\mathcal{F}$ be an r -uniform set system with the property that $F' \cap F'' \not\subseteq F$ for any distinct $F, F', F'' \in \mathcal{F}$. Prove that $|\mathcal{F}| \leq 1 + \binom{r}{\lfloor r/2 \rfloor}$ (Hint: let $F'' \in \mathcal{F}$ be arbitrary and consider the set system $\{F \cap F'' : F \in \mathcal{F} \setminus \{F''\}\}$) [2].

The set system $\mathcal{F}''$ as suggested in the hint is an antichain: for distinct $F, F' \in \mathcal{F}$, having $F' \cap F'' \subseteq F \cap F''$ would in particular imply $F' \cap F'' \subseteq F$ which we assumed not to be the case. Moreover, this same argument shows $|\mathcal{F}''| = |\mathcal{F} \setminus \{F''\}|$, i.e. that $F \cap F'' \neq F' \cap F''$ for distinct F, F' . Since $\mathcal{F}''$ has ground set F'' of size r , it follows from Sperner's Theorem that $|\mathcal{F}''| \leq \binom{r}{\lfloor r/2 \rfloor}$, and hence

$$|\mathcal{F}| = 1 + |\mathcal{F} \setminus \{F''\}| \leq 1 + \binom{r}{\lfloor r/2 \rfloor}.$$

- 1.5 Prove ?? (Hint: try proving the weaker lower bound of n^{-2^d} to get a feel for the proof idea) [2].

The case $d = 1$ is Dilworth's Theorem. Inductively assume we have proven the result up to some value $d \geq 2$. Let G be a graph and $P_1, \dots, P_d$ posets on $V(G)$ such that $x \sim y$ iff $x <_i y$ or $y <_i x$ for some i . If P_d contains a chain on at least $n^{1/(d+1)}$ elements then this corresponds to a clique of size at least $n^{1/(d+1)}$ in G . By the Corollary of Dilworth's Theorem that $n \leq h(P)w(P)$ for all P , we can assume P_d contains an antichain on at least $n^{d/(d+1)}$ vertices of G . Let $G' \subseteq G$ be the induced subgraph of G on this antichain. Since G' is induced on an antichain of P_d , it follows that $P_1, \dots, P_{d-1}$ are such that two vertices of G' are adjacent if and only if they are comparable by some P_i with $1 \leq i \leq d-1$, so inductively G' has either a clique or independent set of size at least $(n^{d/(d+1)})^{1/d} = n^{1/(d+1)}$, proving the result.

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1.7a *Prove Erdős-Szekeres using Dilworth's Theorem [2].*

Construct a poset P on $[n]$ where $i <_P j$ if and only if $a_i < a_j$ and $i < j$ (with these latter inequalities being with respect to the usual ordering on $\mathbb{R}$). By Dilworth there exists a chain or an antichain of size at least $\sqrt{n}$ of P , and it is straightforward to check that these correspond to increasing or decreasing subsequences of π of the same size.

1.7b *Prove this without using Dilworth's Theorem (Hint: for each j define i_j to be the length of the longest increasing subsequence which ends at a_j and d_j to be the length of the longest decreasing subsequence which ends at a_j . Can we have $(i_j, d_j) = (i_{j'}, d_{j'})$ for $j < j'$?) [2+].*

Following the hint, we can not have $(i_j, d_j) = (i_{j'}, d_{j'})$ for $j < j'$: if $a_j < a_{j'}$ then we could add $a_{j'}$ to the end of the increasing subsequence of length i_j ending at a_j to find $i_{j'} > i_j$; an analogous argument works if $a_j > a_{j'}$. It follows that the n pairs (i_j, d_j) are distinct, which means we can not have $i_j, d_j < \sqrt{n}$ for all n and hence some i_j or d_j is at least this large, corresponding to a large increasing or decreasing subsequence.

1.8 *Prove Mirsky's Theorem: if $P = (X, \leq)$ is a finite poset of height $h(P)$, then there exists a partition of X into $h(P)$ antichains (Hint: prove that the set of maximal elements M of P form an antichain) [2].*

For the hint, if two maximal elements x, y were comparable, say with $x < y$, then x would not be maximal. We conclude that M is indeed an antichain.

We now prove the main result by induction on $h(P)$, the case $h(P) = 0$ being trivial since this can only occur if $P = \emptyset$. We claim that $P - M$ has height at most $h(P) - 1$; indeed, if $P - M$ had a chain of length $h(P)$ then whatever element is at the top of the chain is not maximal and hence could be extended to a chain of length $h(P) + 1$ in P , a contradiction. Inductively then we can decompose $P - M$ into at most $h(P) - 1$ antichains, which together with M gives the desired partition.